

HDOC DAY : 21

AIM To write a menu-driven C program using a switch statement and loops to take two positive integers m and n ($m \leq n$) and perform operations like printing multiplication, addition, subtraction, and division tables until the user chooses to exit.

ALGORITHM

1. **Start**
2. Take two positive integers m and n from the user. Check if $m \leq n$. If not, ask the user to enter valid values.
3. Display the menu options:
 - a. Option 1: Multiples of each number
 - b. Option 2: Addition table
 - c. Option 3: Subtraction table
 - d. Option 4: Division table (rounded to 2 decimal places)
 - e. Option 5: Exit
4. Read the user's menu choice.
5. Using a switch statement:
 - a. **Case 1:** Run nested loops to print the first 10 multiples for numbers from m to n.
 - b. **Case 2:** Run nested loops to add numbers from 1 to 10 for each integer from m to n.
 - c. **Case 3:** Run nested loops to subtract numbers from 1 to 10 for each integer from m to n.

- d. **Case 4:** Run nested loops to divide each integer from m to n by numbers from 1 to 10 and display the result formatted to 2 decimal places (%.2f).
 - e. **Case 5:** Exit the program loop.
 - f. **Default:** Print "Invalid choice! Please try again."
6. Repeat steps 3 to 5 using a do-while loop until the user chooses Option 5.
7. Stop

TEST CASE TABLE

TEST CASE	Input m	Input n	Menu choice	Expected output
1	2	4	1	Prints first 10 multiples for numbers 2, 3, and 4
2	2	4	2	Prints addition table (+1 to +10) for numbers 2, 3, and 4
3	2	4	3	Prints subtraction table (-1 to -10) for numbers 2, 3, and 4
4	2	4	4	Prints division table

				(/1 to /10) for numbers 2, 3, and 4 up to 2 decimals
5	2	4	8	Prints "Invalid Choice! Try again."
6	2	4	5	Exits the program

C PROGRAM :

```
#include <stdio.h>

int main() {
    int m, n;
    int choice;
    int i, j;

    printf("Enter starting number (m): ");
    scanf("%d", &m);
    printf("Enter ending number (n): ");
    scanf("%d", &n);

    if (m > n || m <= 0 || n <= 0) {
        printf("Invalid input! Make sure m <= n and both are positive.\n");
        return 0;
    }

    do {
        printf("\n--- MENU ---\n");
        printf("1. Print first 10 multiples\n");
        printf("2. Print addition table\n");
        printf("3. Print subtraction table\n");
        printf("4. Print division table\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("\n--- MULTIPLES ---\n");
                for (i = m; i <= n; i++) {
                    printf("Multiples of %d:\n", i);
                    for (j = 1; j <= 10; j++) {
                        printf("%d x %d = %d\n", i, j, i * j);
                    }
                    printf("\n");
                }
                break;

            case 2:
                printf("\n--- ADDITION TABLE ---\n");
                for (i = m; i <= n; i++) {
                    printf("Addition table for %d:\n", i);
                    for (j = 1; j <= 10; j++) {
                        printf("%d + %d = %d\n", i, j, i + j);
                    }
                    printf("\n");
                }
                break;
        }
    } while (choice != 5);
}
```

```

        case 3:
            printf("\n--- SUBTRACTION TABLE ---\n");
            for (i = m; i <= n; i++) {
                printf("Subtraction table for %d:\n", i);
                for (j = 1; j <= 10; j++) {
                    printf("%d - %d = %d\n", i, j, i - j);
                }
                printf("\n");
            }
            break;

        case 4:
            printf("\n--- DIVISION TABLE ---\n");
            for (i = m; i <= n; i++) {
                printf("Division table for %d:\n", i);
                for (j = 1; j <= 10; j++) {
                    printf("%d / %d = %.2f\n", i, j, (float)i / j);
                }
                printf("\n");
            }
            break;

        case 5:
            printf("Exiting program...\n");
            break;

        default:
            printf("Invalid choice! Please enter a valid option.\n");
    }
} while (choice != 5);

return 0;
}

```

OUTPUT EXECUTION SCREENSHOT :

```
Last login: Mon Sep 14 11:59:32 on console
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % nano 21q1.c
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % gcc 21q1.c -o 21q1
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./21q1
Enter starting number (m): 2
Enter ending number (n): 4

--- MENU ---
1. Print first 10 multiples
2. Print addition table
3. Print subtraction table
4. Print division table
5. Exit
Enter your choice: 1

--- MULTIPLES ---
Multiples of 2:
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20

Multiples of 3:
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30

Multiples of 4:
4 x 1 = 4
4 x 2 = 8
4 x 3 = 12
4 x 4 = 16
4 x 5 = 20
4 x 6 = 24
4 x 7 = 28
4 x 8 = 32
4 x 9 = 36
4 x 10 = 40
```

```
--- MENU ---  
1. Print first 10 multiples  
2. Print addition table  
3. Print subtraction table  
4. Print division table  
5. Exit  
Enter your choice: 2
```

```
--- ADDITION TABLE ---  
Addition table for 2:  
2 + 1 = 3  
2 + 2 = 4  
2 + 3 = 5  
2 + 4 = 6  
2 + 5 = 7  
2 + 6 = 8  
2 + 7 = 9  
2 + 8 = 10  
2 + 9 = 11  
2 + 10 = 12
```

```
Addition table for 3:  
3 + 1 = 4  
3 + 2 = 5  
3 + 3 = 6  
3 + 4 = 7  
3 + 5 = 8  
3 + 6 = 9  
3 + 7 = 10  
3 + 8 = 11  
3 + 9 = 12  
3 + 10 = 13
```

```
Addition table for 4:  
4 + 1 = 5  
4 + 2 = 6  
4 + 3 = 7  
4 + 4 = 8  
4 + 5 = 9  
4 + 6 = 10  
4 + 7 = 11  
4 + 8 = 12  
4 + 9 = 13  
4 + 10 = 14
```

```
--- MENU ---
1. Print first 10 multiples
2. Print addition table
3. Print subtraction table
4. Print division table
5. Exit
Enter your choice: 3
```

```
--- SUBTRACTION TABLE ---
Subtraction table for 2:
2 - 1 = 1
2 - 2 = 0
2 - 3 = -1
2 - 4 = -2
2 - 5 = -3
2 - 6 = -4
2 - 7 = -5
2 - 8 = -6
2 - 9 = -7
2 - 10 = -8
```

```
Subtraction table for 3:
3 - 1 = 2
3 - 2 = 1
3 - 3 = 0
3 - 4 = -1
3 - 5 = -2
3 - 6 = -3
3 - 7 = -4
3 - 8 = -5
3 - 9 = -6
3 - 10 = -7
```

```
Subtraction table for 4:
4 - 1 = 3
4 - 2 = 2
4 - 3 = 1
4 - 4 = 0
4 - 5 = -1
4 - 6 = -2
4 - 7 = -3
4 - 8 = -4
4 - 9 = -5
4 - 10 = -6
```

```
--- MENU ---
1. Print first 10 multiples
2. Print addition table
3. Print subtraction table
4. Print division table
5. Exit
Enter your choice: 4
```

```
--- DIVISION TABLE ---
```

```
Division table for 2:
```

```
2 / 1 = 2.00
2 / 2 = 1.00
2 / 3 = 0.67
2 / 4 = 0.50
2 / 5 = 0.40
2 / 6 = 0.33
2 / 7 = 0.29
2 / 8 = 0.25
2 / 9 = 0.22
2 / 10 = 0.20
```

```
Division table for 3:
```

```
3 / 1 = 3.00
3 / 2 = 1.50
3 / 3 = 1.00
3 / 4 = 0.75
3 / 5 = 0.60
3 / 6 = 0.50
3 / 7 = 0.43
3 / 8 = 0.38
3 / 9 = 0.33
3 / 10 = 0.30
```

```
Division table for 4:
```

```
4 / 1 = 4.00
4 / 2 = 2.00
4 / 3 = 1.33
4 / 4 = 1.00
4 / 5 = 0.80
4 / 6 = 0.67
4 / 7 = 0.57
4 / 8 = 0.50
4 / 9 = 0.44
4 / 10 = 0.40
```

```
--- MENU ---
```

```
1. Print first 10 multiples
2. Print addition table
3. Print subtraction table
4. Print division table
5. Exit
```

```
Enter your choice: 8
```

```
Invalid choice! Please enter a valid option.
```

```
--- MENU ---
```

```
1. Print first 10 multiples
2. Print addition table
3. Print subtraction table
4. Print division table
5. Exit
```

```
Enter your choice: 5
```

```
Exiting program...
```

```
manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ %
```